Game Design

Flow

Main menu → **world map** (scroll the 8 realms, tap an unlocked level) → **story briefing** (characters talk, tap to advance, Skip) → level → result → Next goes to the next story.

Core loop (one level, 30–90 s, real time)

1. **Intro:** the Vanguard soldiers march in from the left and line up behind the catapult. A wooden sign shows the level name. Level 1 shows a "Drag & Release to Fire!" hand tutorial.
2. **Play, all in real time:**
3. Drag on the left part of the screen to pull the catapult back; a dotted line shows the arc. Release to fire.
4. The next hero hops into the bucket `catapult.reload_s + hop_s` (≈0.7 s) after a shot. You can fire again while the last shot is still flying or the tower is still falling.
5. Tap (short touch, no drag) to use the ability of the newest hero still in the air.
6. Every archer shoots on its own timer (`archers.fire_min_s..fire_max_s`), pulls its bow first, and stops shooting while it is falling or knocked over.
7. **Win:** all archers defeated → slow motion, soldiers charge the fortress, stars. **Lose:** all soldiers knocked out, or no heroes left and the fortress has settled.

Our side

- **Catapult**: wooden throwing arm on wheels. Our own design (not a slingshot). The arm pulls back when dragged, and the projectile sits in the bucket.
- **Soldiers** (the Vanguard): 4 round soldiers stand behind the catapult. Each has 3 HP, loses 1 per arrow hit, and blocks 30% of arrows with the shield. At 0 HP a soldier sits down dazed ("knocked out", no gore). They march in, cheer on good hits and charge when you win.
- **Hero loadout:** before each battle, the Wizard's Arsenal lets the player choose exactly four heroes and arrange their firing order. Heroes unlock through the campaign story. Per-hero caps and an eight-point arcane-power budget prevent four-shot specialist spam. Recommendations and saved per-level loadouts remain available.

Projectiles (heroes)

Values live in `data/projectiles.json`; the ones below are starting points.

Hero	Mass	Special (tap in flight)	Strong vs	Weak vs
Rocky (stone)	heavy	Slam: dives straight down with 2.5× mass	stone, archers	none
Blaze (fire pot)	light	Fire dash: speeds up 1.55×. Always bursts into flames on impact (bigger burst after a dash); burning wood loses HP and spreads to touching wood	wood, roofs	stone
Boomer (bomb)	medium	Explodes (also explodes 1.5 s after first impact if not tapped). Radial impulse + damage, radius 220.	everything nearby	far targets

Hero	Mass	Special (tap in flight)	Strong vs	Weak vs
Twister (cyclone)	light	Turns into a whirlwind for 2.5 s that drifts forward, pulls nearby light bodies in and lifts them, then flings them.	light wood, archers on ledges	heavy stone
Quaky (quake stone)	very heavy	Earthbreak: dives and sends a wide ground shockwave through low supports; marked weak points take extra damage.	foundations, weak points	high targets

Enemies

- **Redcap Archer**: stands on fortress floors. 30 HP. Shoots a ballistic arrow at a random living soldier every 5.5–8.5 s, with aim error ±spread (data). Variants later: Shield Archer (blocks the front), Captain (fires 3 arrows), Fire Archer (burning arrows).
- Archers fall over and are defeated at 0 HP, or when they drop off the map.

Materials

Values live in `data/materials.json`.

Material	Look	HP	Density	Damage taken ×	Flammable
wood plank	warm brown planks	60	0.6	1.0	yes
wood crate	crate with cross brace	45	0.5	1.15	yes
stone	grey rounded bricks	140	1.6	0.35	no
iron	dark riveted plate	260	2.4	0.22	no
crystal	glowing faceted crystal	40	0.9	1.35	no, shatters and hurts blocks within 120 px
boulder	round granite (rolls)	400	2.2	0.2	no
log	round trunk on slopes (rolls, rams what it hits)	600	0.8	0.15	yes
wheel	big cart wheel (level 35: rolls down and rams one part of the fort)	1200	1.6	0.12	no
straw roof	thatch triangle	35	0.3	1.3	very
powder barrel	red barrel	25	0.6	1.6	explodes
ice	pale translucent blue, frosty top	40	0.9	1.3	no, melts under fire (32 HP/s), slippery, shatters
rune crystal	violet socket with a glowing gem and an arrow	150	1.2	0.45	no, absorbs energy and fires a beam

Iron beams bend into a V on a hard hit, and fire makes iron red-hot for 5 s (×2.5 damage). See `docs/SIGNATURE_FEATURES.md` §1 for the full rules.

Each hero also has a `material_mult` per material (for example the bomb is better against iron). Each block shows 3 damage stages (intact, cracked, badly cracked) and breaks into debris pieces.

Castle mechanics

- Marked weak blocks have reduced HP, take extra damage, and can trigger a gate, drawbridge or captive release.
- Gates, drawbridges with visible pulley/counterweight rigs, moving platforms, pendulum obstacles, spikes and lava are data-driven level pieces.
- Captives in levels 6 and 12 must be rescued as well as defeating every archer.
- **Rope loads**: boulders, crates or cages hang from ropes. Any hero touching a rope, fire within 120 px or an explosion cuts it and the load drops; it can also trigger another mechanism.
- **Hanging platforms** (gondolas and rope bridges) hang from two ropes: one cut tips them, two cuts drop them. Drawbridges show their working pulley and counterweight.
- **Flood gates** release a timed wave that pushes and floats blocks and drowns covered guards. **Water wheels** wait for a brake trigger, then their geared paddles sweep nearby structures.
- **Breakable ledges** tip and drop their entire load when their keystone trigger breaks.
- **Heavy throwing beams** launch a held boulder or powder keg when the short arm is slammed. **Explosive carts** carry bolted powder barrels down rails after their brake rope is cut.
- **Crushers** slam down on a cycle or when a trigger fires. **Geysers** burst up and throw heroes and pieces; lava geysers also burn.
- **Bolts** pin two blocks together. They snap after 40 damage, in an explosion or quake, or when one block breaks, so bolted towers fall as one piece.
- **Wind** (levels 29, 41–43, 45, 47–50) gusts left or right and pushes heroes in flight; the HUD shows it. **Tide** (levels 21, 23, 26, 28, 30) raises and lowers the sea and floating decks.
- **Bosses** (every 10th level): a crowned archer at 1.55× size with 160 HP and 3-arrow volleys. At HP thresholds it changes phase: reinforcements drop in, a 4 s shield, faster fire (enrage), or its moving castle speeds up 1.7× (charge). Defeat: 15,000 points. HUD boss bar with phase pips.
- Hero unlocks: Rocky 1, Blaze 3, Boomer 6, Twister 12, Quaky 22.
- Realm weather adds petals, leaves, pollen, rain or embers over the five painted environments.
- The fortress stands on bases across a gap from the catapult ledge. Anything knocked into the gap splashes and is gone: blocks count as destroyed (scored), archers as defeated.

Story scenes

No text screens. Each story is an animated scene played in the level: characters walk or drop in, talk in speech bubbles, react, and the camera points out the new mechanic with a ring, arrow and sign. New mechanics get a ghost-shot demo; new heroes get a summon and ability preview; bosses get a roar and name plate. Every level opens with a short flyover with up to 3 callouts. Tap skips a beat, SKIP skips the scene.

Damage

`impulse = contact relative normal speed × reduced mass`. Below `min_impulse` = no damage. `damage = impulse × source_factor × material_factor`, with a per-body cooldown of 0.1 s. Source factor: projectile 1.0, block 0.5, ground 0.4 (only above `ground_min_speed`). Settling contacts never deal damage. Physics starts in "sleep" for the first 1.0 s so towers settle.

Scoring and stars

- Archer defeated: 5,000. Block destroyed: 500 (stone 800). Unused hero: 10,000.
- Soldier knocked out: –2,000.
- Stars from the level's `star_scores` thresholds `[1★, 2★, 3★]` in the level file.

Economy

- **Coins**: earned per level (stars × 50, first clear only + 10 on replays) and from rewarded ads.
- **Wizard's Workshop**: 3 upgrade levels per hero for 150 / 350 / 700 coins (Rocky +15% slam, Blaze +14% burst, Boomer +12% blast, Twister +18% whirl time, Quaky +12% shockwave reach).
- **Spend**: extra hero mid-level (+1 bomb, 150 coins), soldier shield upgrade, catapult skins.
- **Gems**: not in v1. One currency keeps it simple.

Energy (army stamina)

- Up to `energy.max` (10) shield bars. Refill is non-linear: `energy.refill_s` = 1:00, 1:30, 2, 2:30, 3, 3:20, 3:40, 4, 4:20, 4:40 min per bar (the first missing bar comes back fastest; empty to full is 30 min, a test keeps it at or under 30). When all bars are back the "energy full" reminder fires (Notifications). Refill runs on the wall clock from the saved time, so it continues while the app is closed.
- Levels up to `energy.free_until` (5) are free. A level costs 1 bar, 2 from realm 3, 3 for a boss. Retries cost too.
- Short on bars: the "YOUR ARMY IS EXHAUSTED!" popup offers an opt-in rewarded ad that pays for that one level only. The next level after an ad-paid one needs rested bars. Training ranges never cost energy.
- Code: `src/game/energy.gd` (rules, state in `Save.data.energy`), `src/ui/energy_ui.gd` (pill, popup).

Training ranges

- Before the first level of a new hero (fire before 3, bomb 6, cyclone 12, quake 22) a short range shows how to use it. Rocky has a range too, reachable only from the Workshop. Tap a hero card in Master Orin's Workshop to practise again.
- Source: `TRAINING` in `tools/campaign.py`, generated to `data/training/`. Coach texts in `balance.training`.

Rocky addons

- Hats (pirate hat, cap, crown, viking helmet, top hat) and glasses (shades, specs), one per slot, sold as Play one-time products. Worn on Rocky everywhere he appears (catapult, queue, flight, home). Looks only.
- Code: `src/game/hero_addons.gd`; catalogue `balance.cosmetics.hero_addons`; art `assets/art/svg/addon_*.svg`.

Upgrade prices

- All Workshop upgrade tracks (Vanguard and each siege engine) cost twice the earlier prices. Forge (unlock) prices are unchanged.

Profile name

- Default `balance.profile.default_name`. Rename is free once level `free_rename_level` (5) is reached, else 20 gems.
- The rename popup uses an in-game keyboard on the lower screen (the system keyboard floated over the field in landscape).

Controls

- Drag anywhere on the left third to aim (not only on the catapult, easier on phones).
- Pinch to zoom, two-finger drag to pan while nothing is in flight.
- Tap to use ability. Pause button top left. Restart button next to it.

Feel ("juice") checklist

Screen shake scaled by impulse, 60 ms hit-stop on big impacts, dust puffs, wood splinters and stone chips, squash on launch and landing, slow motion when the last archer falls, star pop animation, coin fly-to-counter, haptic tap on launch and big hits (Android vibrate, setting to turn off).